

Experiment - 01

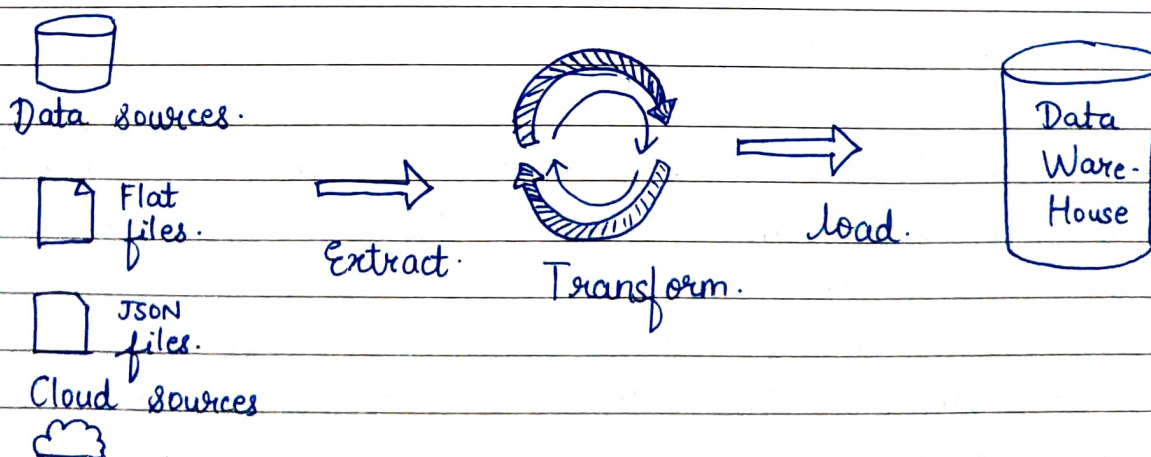
- Aim: Import the legacy data from different sources such as (Excel, Sql server, Oracle, etc) and load in the target system (You can download sample database such as Adventure works, Northwind, foodmart, etc).

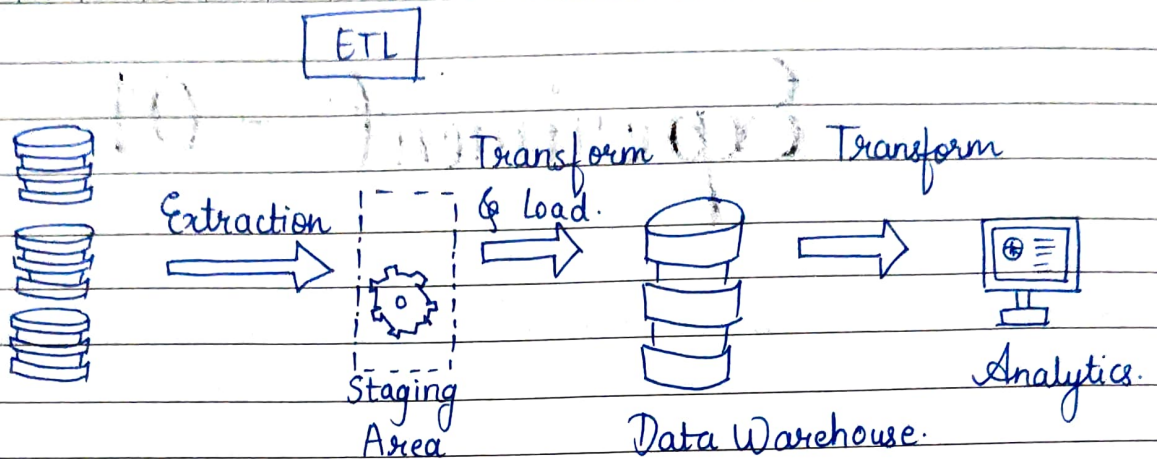
- Theory:

(i) Introduction -

In modern enterprises, data resides in multiple legacy systems such as excel files, SQL server databases, Oracle databases, ERP systems and flat files. These systems are typically designed for transaction processing and not for analytical processing. To enable decision support systems (DSS) and Business Intelligence (BI), data must be consolidated into a centralized analytical repository such as a data warehouse. This is achieved through the ETL.

ETL = Extract + Transform + Load.





(2) Architecture Components -

1. Source systems
 - Excel files.
 - SQL Server database.
 - Oracle database.
2. Staging Area.
 - Temporary storage.
 - Raw extracted data stored without heavy transformation.
3. ETL Engine.
 - Data cleansing.
 - Data transformation.
 - Data validation.
4. Target System.
 - Data warehouse.
 - Data Mart.
 - BI reporting system.

(3) Foundations of Data Migration -

- Data Quality.
- Data Integrity.
- Data consistency.

Organizations modernize systems to improve scalability, performance, and analytics. However legacy systems may contain:

- Inconsistent formats.
- Redundant records.
- Data quality issues.
- Missing constraints.

Thus migration must ensure,

$$\text{Data}_{\text{target}} = f(\text{Data}_{\text{source}})$$

Where, f is extraction, transformation, validation, loading function.

(4) Data Migration Phases -

Phase 1 - Data Extraction.

Sources - Excel (.xls, .xlsx, .csv), SQL server, Oracle, Flat files, APIs.

Extraction Methods -

<u>Source</u>	<u>Method</u>
Excel	OleDb / CSV parser
SQL Server	SELECT queries, SSIS
Oracle	SQL *Loader, ODBC
APIs.	Rest calls.

Phase 2: Data Transformation.

Transformation ensures:

- Data type consistency
- Data cleansing.
- Deduplication.
- Business rule application.
- Format standardization.

Phase 3: Data Loading

(1) Full Load: Entire dataset loaded at once.

Total rows loaded = Source row count.

(2) Incremental load. - Only new or changed records loaded.

Incremental rows = Source rows - Previously loaded rows.

(5) Data Migration strategies -

1. Big Bang Migration.

- Entire data migrated at once.
- System downtime required.
- High risk.

2. Trickle Migration.

- Data Migrated in phases.
- Low risk.
- Parallel run possible.

(6). Example using sample databases -

Using adventureworks :

- Extract sales data from SQL server.
- Transform currency formats.
- Load into data warehouse.

Using Northwind:

- Migrate customers and orders from legacy systems.
- Clean invalid postal codes.
- Normalize product categories.

(7). Data validation Techniques.

→

7.1 Row count validation.

$$\text{Source count} = \text{Target count.}$$

7.2. Checksum Validation.

$$\text{Checksum} = \sum \text{Hash (row)}$$

7.3 Reconciliation formula.

$$\text{Variance} = \text{Source - Total} - \text{Target - Total.}$$

If, variance = 0,

→ We can conclude migration was successful.

(8) Tools used for Migration -

1. SSIS - SQL Server ETL.
2. Oracle data integrator - Oracle ETL.
3. Talend - Open source ETL.
4. Informatica - Enterprise ETL.
5. Python (Pandas) - Custom Migration.
6. Azure data factory - Cloud ETL.

(9) Best Practices -

- Perform data profiling before migration.
- Use staging tables.
- Maintain audit logs.
- Backup legacy data.
- Perform multiple test runs.
- Validate with business users.
- Document source to target mappings.

• CONCLUSION:

Importing legacy data from Excel, SQL server and Oracle into a centralized BI system is essential for modern enterprises. The ETL process transforms scattered operational data into structured analytical repositories.

Using Power BI -

- Extract → get data.
- Transform → Power Query.
- Load → Data model.
- Analyze → DAX.
- Visualize → Dashboard.

This completes the BI lifecycle from raw legacy data to intelligent decision making.

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